

Business Problem

The stock market generates a massive volume of trading data every day, making it difficult to identify meaningful trends and investment opportunities through raw data alone. Investors and analysts need a clear way to determine which dates experienced the highest trading activity, which stocks were traded the most, and how trading volume varies across different days of the week. Additionally, understanding stock price volatility, especially for companies like Amazon (AMZN), is essential for assessing market risk. There is also a need to evaluate historical stock performance to identify which investment would have produced the highest return between 2014 and 2017.

The goal of this analysis is to:

- Which date in the sample saw the largest overall trading volume?
- On which day of the week does volume tend to be highest? Lowest?
- On which date did Amazon (AMZN) see the most volatility, measured by the difference between the high and low price?
- If you could go back in time and invest in one stock from 1/2/2014-12/29/2017, which would you choose? What % gain would you realize?

• Analysis Insights

Summary Statistics

	date	open	high	low	close	volume
count	497461	497461.0	497461.0	497461.0	497461.0	497461.0
mean	2016-01-06 17:16:56.524310	86.0	87.0	86.0	86.0	4253695.0
min	2014-01-02 00:00:00	2.0	2.0	2.0	2.0	101.0
25%	2015-01-08 00:00:00	42.0	42.0	41.0	42.0	1080183.0
50%	2016-01-11 00:00:00	65.0	66.0	64.0	65.0	2085013.0
75%	2017-01-06 00:00:00	98.0	99.0	98.0	98.0	4271999.0
max	2017-12-29 00:00:00	2044.0	2068.0	2035.0	2049.0	618237630.0
std	NaN	101.0	102.0	101.0	101.0	8232210.0

Dataset Size: The dataset contains 497,461 records, providing a large amount of market data for analysis.

Price Statistics: The average open and close prices are around 86, while the median close price is 65, indicating variation in prices.

Price Range: Prices range from a minimum of 2 to a maximum close of 2,049, showing significant price variation over time.

Trading Volume: Average volume is approximately 4.25 million, while the maximum reaches about 618.24 million, indicating major volume spikes.

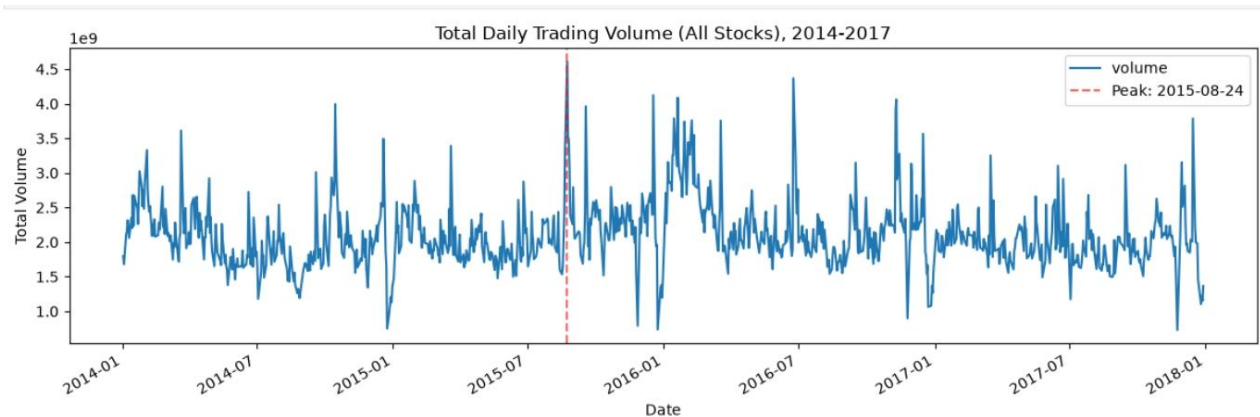
Overall Insight: The data covers 2014–2017 and shows substantial changes in both stock prices and trading activity, making it useful for trend and market-volume analysis.

Questions & Key Findings

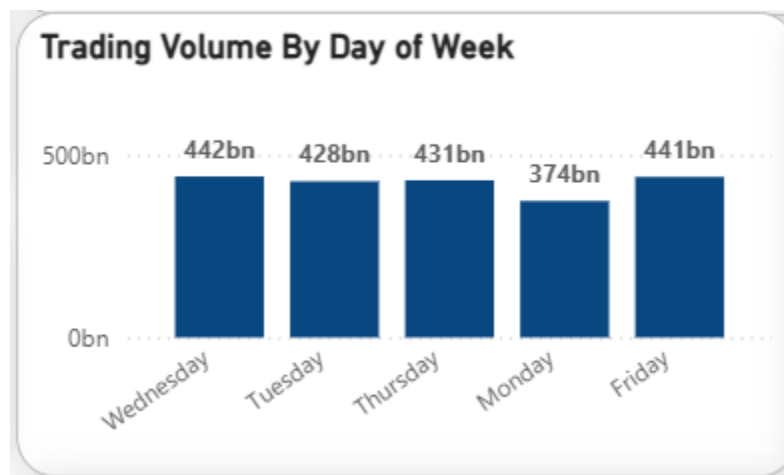
1. Which date saw the largest overall trading volume?

date	
2015-08-24	4607945196
2016-06-24	4367393052
2015-12-18	4124454411
2016-01-20	4087629753
2016-11-10	4060601612
2014-10-15	3993171524
2015-09-18	3962050449
2016-11-09	3915089371
2016-01-15	3787884056
2017-12-15	3786992731

- **Highest total trading volume date is 2015-08-24**

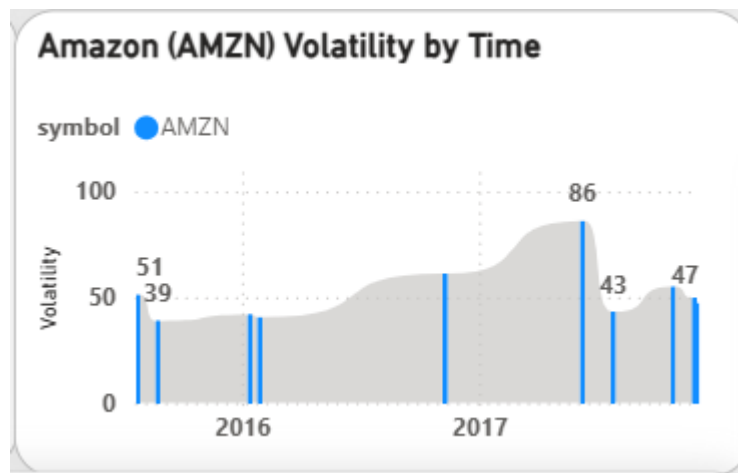


2. On which day of the week is volume highest / lowest, on average?



- **Highest Volume:** Wednesday has the highest average trading volume at approximately 442 billion.
- **Second Highest:** Friday follows closely with around 441 billion, indicating strong trading activity.
- **Moderate Volume:** Thursday and Tuesday record approximately 431 billion and 428 billion, respectively.
- **Lowest Volume:** Monday has the lowest average trading volume at about 374 billion.
- **Overall Insight:** Trading activity is generally higher from Tuesday to Friday, while Monday shows the lowest average volume.

3. On which date did Amazon (AMZN) see the most volatility (high – low)?

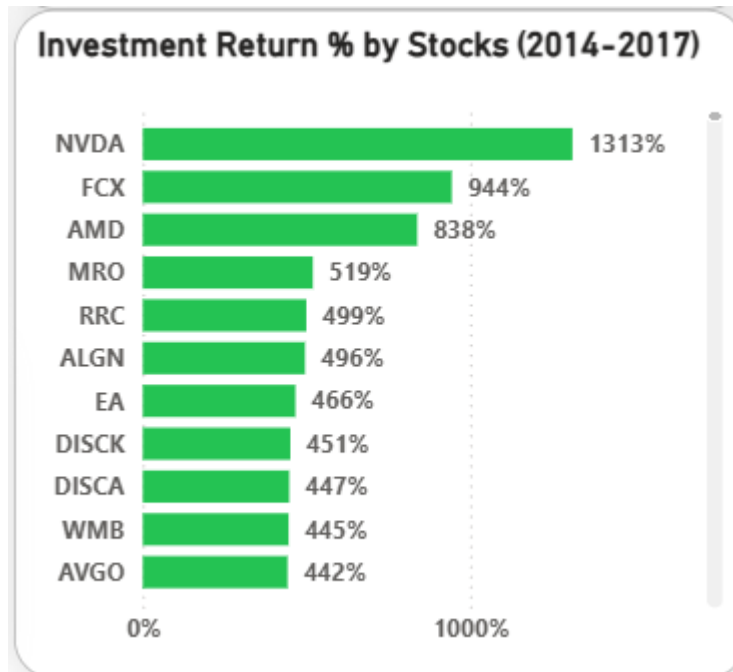


- **Volatility Trend:** Amazon (AMZN) shows noticeable fluctuations in volatility between 2015 and 2017, with several sharp spikes.
- **Highest Volatility:** The highest recorded volatility is approximately 85.99 on 9 June 2017.
- **Other Peaks:** Earlier volatility spikes reach around 51, while several observations remain between 35–50.
- **Overall Insight:** The large fluctuations indicate periods of increased market uncertainty and significant price movement in Amazon's stock.

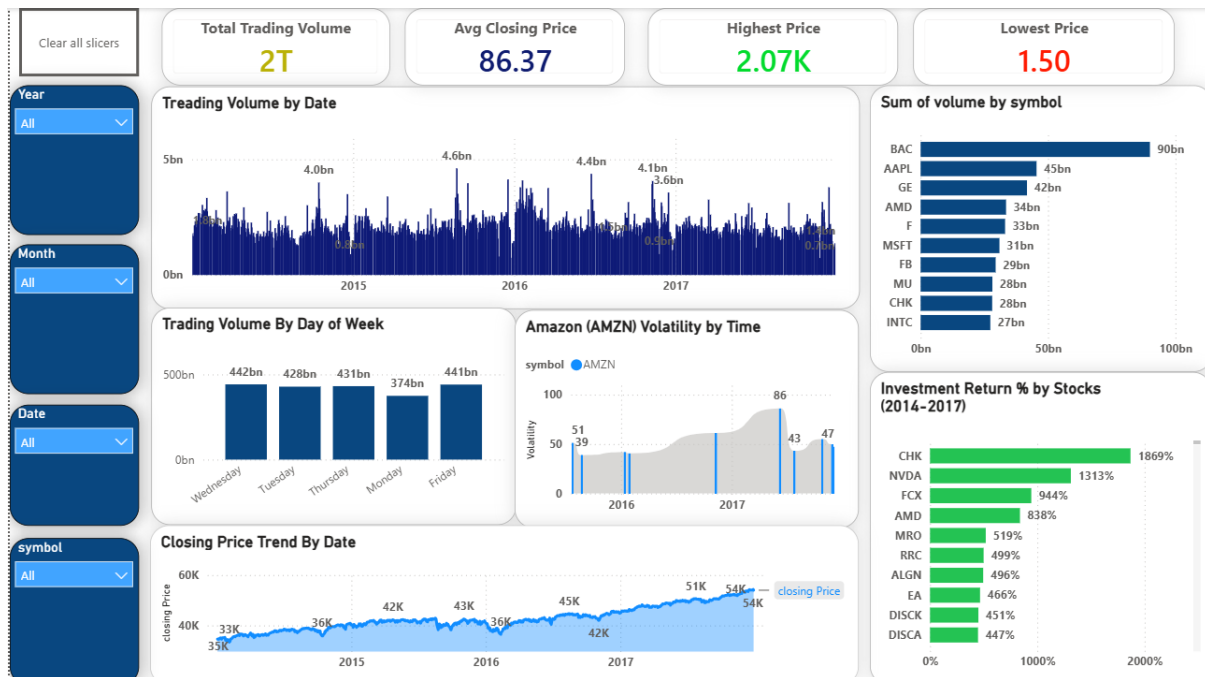
4. Which single stock, bought on 1/2/2014 and held to 12/29/2017, would have delivered the best return?

- **Highest Return:** CHK generated the highest investment return at 1,869%, making it the top-performing stock.
- **Strong Performers:** NVDA and FCX achieved returns of 1,313% and 944%, respectively.
- **Moderate Returns:** AMD recorded 838%, followed by MRO at 519% and RRC at 499%.
- **Other Stocks:** ALGN, EA, DISCK, and DISCA delivered returns between 447% and 496%.

- **Overall Insight:** All listed stocks generated positive returns from 2014–2017, with NVDA significantly outperforming the other stocks.



Overall Analysis



- **Overall Performance:** The dashboard shows strong growth in stock prices from 2014–2017, with the average closing price at 86.37 and the highest price reaching approximately 2.07K.
- **Trading Activity:** Total trading volume is around 2T, with BAC recording the highest symbol-wise volume at about 90bn.
- **Volume by Day:** Wednesday has the highest average trading volume (442bn), while Monday has the lowest (374bn).
- **Volatility & Returns:** Amazon shows volatility spikes up to 86, while CHK delivers the highest investment return at 1,869%, followed by NVDA at 1,313%.
- **Key Insight:** Overall, the analysis indicates strong market growth, high trading activity, and significant differences in stock performance and volatility, providing useful insights for investment and risk assessment.